

Full Stack Web Application Roadmap & Architecture

This document provides a professional roadmap and architecture for building a scalable full stack labor management system using React, Node.js, and Prisma.

1. Backend Architecture (Production Style)

```
src
├── app.ts
├── server.ts
├── config
│   └── env.ts
├── routes
│   ├── index.ts
│   ├── auth.routes.ts
│   ├── user.routes.ts
│   ├── site.routes.ts
│   ├── worker.routes.ts
│   ├── payment.routes.ts
│   └── report.routes.ts
├── controllers
│   ├── auth.controller.ts
│   ├── user.controller.ts
│   ├── site.controller.ts
│   ├── worker.controller.ts
│   ├── payment.controller.ts
│   └── report.controller.ts
├── services
│   ├── auth.service.ts
│   ├── user.service.ts
│   ├── site.service.ts
│   ├── worker.service.ts
│   ├── payment.service.ts
│   └── report.service.ts
├── repositories
│   ├── user.repository.ts
│   ├── site.repository.ts
│   ├── worker.repository.ts
│   └── payment.repository.ts
├── middlewares
│   ├── auth.middleware.ts
│   ├── role.middleware.ts
│   └── error.middleware.ts
├── utils
│   ├── jwt.ts
│   ├── password.ts
│   ├── logger.ts
│   └── date.ts
├── validators
│   ├── auth.validator.ts
│   └── user.validator.ts
├── prisma
│   └── client.ts
```

2. Frontend Architecture (React Production Structure)

```
src
├── main.tsx
├── App.tsx
├── api
│   ├── axios.ts
│   ├── auth.api.ts
│   ├── site.api.ts
│   └── worker.api.ts
```

```

■    ■■■ payment.api.ts
■
■    ■■■ hooks
■    ■■■ useAuth.ts
■    ■■■ useFetch.ts
■
■    ■■■ context
■    ■■■ AuthContext.tsx
■
■    ■■■ layouts
■    ■■■ DashboardLayout.tsx
■    ■■■ AuthLayout.tsx
■
■    ■■■ components
■    ■■■ ui
■    ■■■ Button.tsx
■    ■■■ Table.tsx
■    ■■■ Card.tsx
■    ■■■ Modal.tsx
■
■    ■■■ charts
■    ■■■ MonthlyExpenseChart.tsx
■    ■■■ WorkerPaymentChart.tsx
■
■    ■■■ forms
■    ■■■ LoginForm.tsx
■    ■■■ WorkerForm.tsx
■    ■■■ SiteForm.tsx
■
■    ■■■ pages
■    ■■■ auth
■    ■■■ LoginPage.tsx
■    ■■■ ForgotPasswordPage.tsx
■
■    ■■■ owner
■    ■■■ Dashboard.tsx
■    ■■■ Sites.tsx
■    ■■■ Workers.tsx
■    ■■■ Payments.tsx
■    ■■■ Reports.tsx
■
■    ■■■ foreman
■    ■■■ Dashboard.tsx
■    ■■■ Workers.tsx
■    ■■■ AddDailyRecord.tsx
■    ■■■ Payments.tsx
■
■    ■■■ worker
■    ■■■ Dashboard.tsx
■
■    ■■■ routes
■    ■■■ AppRoutes.tsx
■    ■■■ ProtectedRoute.tsx
■
■    ■■■ types
■    ■■■ user.ts
■    ■■■ worker.ts
■    ■■■ site.ts
■
■    ■■■ utils
■    ■■■ date.ts
■    ■■■ currency.ts

```

3. Development Roadmap

- 1 Step 1: Setup backend infrastructure (server, express app, prisma client).
- 2 Step 2: Implement authentication (login, JWT, password hashing).
- 3 Step 3: Build user management (create users, roles, permissions).
- 4 Step 4: Implement site management (create and manage construction sites).
- 5 Step 5: Implement worker assignment to sites.
- 6 Step 6: Build the foreman dashboard for managing workers.
- 7 Step 7: Implement attendance/work entry system.
- 8 Step 8: Implement payment management.
- 9 Step 9: Add charts and analytics dashboards.

- 10 Step 10: Implement reports and exporting (PDF/Excel).
- 11 Step 11: Add activity logging for auditing.
- 12 Step 12: Implement notifications and alerts.

4. Standard Request Flow

